

Decision Trees and Random Forests

Objective

Learn and implement tree-based machine learning models for classification using Decision Trees and Random Forests.

Tools Used

- Python
- Pandas
- Scikit-learn
- Matplotlib
- Graphviz

Tasks Performed

1. Loaded and preprocessed the dataset.
2. Trained a Decision Tree Classifier.
3. Visualized the Decision Tree structure.
4. Analyzed overfitting by adjusting tree depth.
5. Trained a Random Forest Classifier.
6. Compared model accuracies.
7. Evaluated feature importances.
8. Performed cross-validation for model evaluation.

Results

- Successfully built and evaluated Decision Tree and Random Forest models.
- Compared classification performance between both algorithms.
- Identified the most important features influencing predictions.
- Observed how controlling tree depth helps reduce overfitting.

Files Included

- Task5.ipynb
- Dataset file
- README.md
- Screenshots (optional)

Conclusion

Decision Trees are simple and interpretable models, while Random Forests generally provide better accuracy and robustness by combining multiple trees. Feature importance analysis and cross-validation help in understanding and validating model performance.